Does a streaming video-LLM know what it has forgotten?

Week 1 briefing — two experiments, one probe validation, one collision check

lenrdpei · 18 August 2026

Qwen3-VL-8B-Instruct on EgoLifeQA · 373 questions · 5,540 + 16,565 scored records

Agenda

Three things happened this week; two of them change what we write

1. **Both experiments are in, and they disagree with the proposal.**
   The predicted signature is absent in both regimes. What replaced it is sharper, and the two runs together say something neither says alone.
2. **The confidence probe was audited and it holds.**
   The alternative definition was computed on the committed records at zero GPU cost. One sensitivity found, located and explained — and it is worth a result of its own.
3. **A directly adjacent benchmark appeared in June.**
   EgoSAT. It takes one novelty claim off us, and independently replicates our central finding on a different dataset and different backbones.

What I need from you

1. Sign-off on retiring the original hypothesis (Part 1).
2. A call on which of four cheap follow-ups to run first (Part 4).
3. How hard to position against EgoSAT (Part 3).

Part 1

What the two experiments say

Two regimes, same backbone, same questions. The prediction failed twice, and the replacement is a better claim.

The design, in one slide

Everything is held fixed except how the context reaches the model

	how context reaches the model	scored records
E1	re-encodes the whole window at every checkpoint	5,540
E2	one ingestion into a bounded KV cache, queried at checkpoints	16,565

Same frozen Qwen3-VL-8B, same 373 EgoLifeQA questions, same frame bank, same analysis code. EgoStream is a plug-in memory manager over the same weights, so the regime comparison needs no bridge condition.

Two measures, read at every checkpoint

accuracy — argmax over the option letters vs gold.

confidence — max softmax over the option-letter logits {A,B,C,D} at the first generated position. No text generated. *Not* a calibrated probability.

Every Δ on the following slides is a **paired within-question contrast**, cluster-bootstrapped over questions (5,000 draws). Levels are weak by construction and are not compared across regimes.

Source: `artifacts/m8_teaser.md`, `artifacts/x9_streaming.md` — both generated from the metrics JSON, both authoritative.

The headline first: the hypothesis we set out to test is dead

Tested twice, in the two regimes, and it failed in both

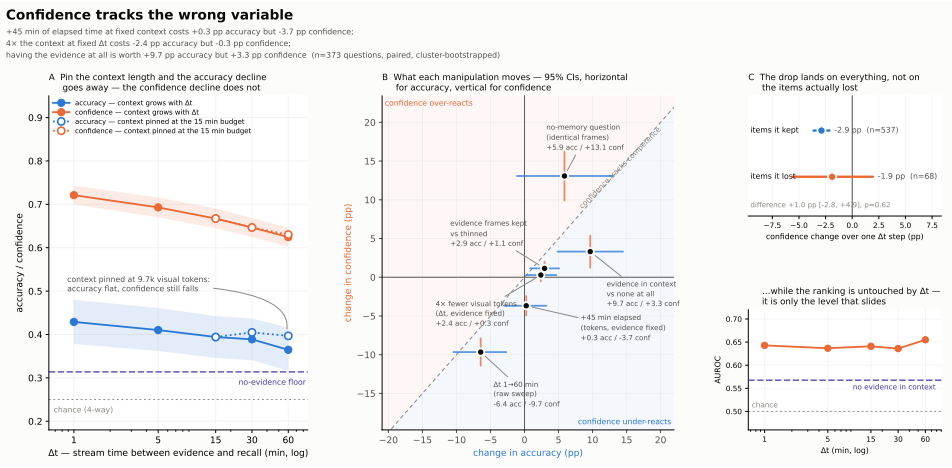
Prediction. The gap between confidence and accuracy *widens* as evidence ages — the model does not know it forgot.

overconfidence gap, Δt $1 \rightarrow 60$ min	change (pp)	p
E1 — full-reprocess	-3.2 $[-7.5, +1.0]$	0.138
E2 — streaming KV cache, evidence genuinely evicted	-4.0 $[-9.0, +1.0]$	0.105

Both point mildly *toward narrowing*. **E2** was the setting the hypothesis most deserved — a bounded cache actually destroys the evidence rather than diluting it — and it still did not appear.

The recommendation is to retire it rather than test it a third way. What the controls found instead is a stronger claim, and the rest of Part 1 is that claim.

E1 (full-reprocess): confidence tracks the wrong variable



Source: `figures/headline.pdf` (M7), cut into panels by `modules/b2_figures.py`. Nothing redrawn.

E1 — the token-matched arms are what make this readable

Elapsed time and context length are confounded in the raw sweep; two extra arms separate them

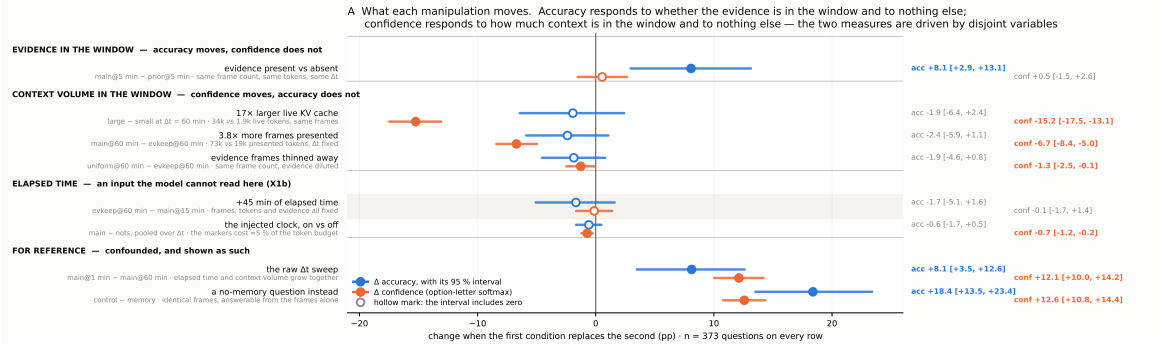
manipulation	Δaccuracy	Δconfidence
+45 min elapsed *(tokens and evidence fixed)*	+0.3 [−2.9, +3.2]	−3.7 [−4.9, −2.4]
4× fewer visual tokens *(Δt and evidence fixed)*	+2.4 [+0.3, +4.7]	+0.3 [−0.5, +1.1]
evidence in context vs none *(token-matched, Δt = 5 min)*	+9.7 [+4.8, +14.5]	+3.3 [+1.2, +5.4]
raw Δt sweep $1 \rightarrow 60$ min *(confounded: both grow together)*	−6.4 [−10.5, −2.4]	−9.7 [−11.5, −7.9]

Confidence moves on elapsed time — the one manipulation that costs *no* accuracy. **Accuracy** moves on context length and on whether the evidence is there at all, and confidence barely registers either: the largest accuracy effect in the experiment is under-reacted to by 2.9×.

Read down the table: the two token-matched arms make elapsed time and context length separately identifiable, and they come apart. This is panel B of the previous figure, in numbers.

E2 (streaming KV cache): the same dissociation, rotated

Accuracy responds to whether the evidence is in the window; confidence, to how much context is in it



The elapsed-time rows are drawn as an *unavailable* axis, not a null — see the slide after next. The whole figure, with its four lower panels, is in the backup.

Source: `figures/streaming_headline.pdf` (X10), drawn from `x9_regime_comparison.json`.

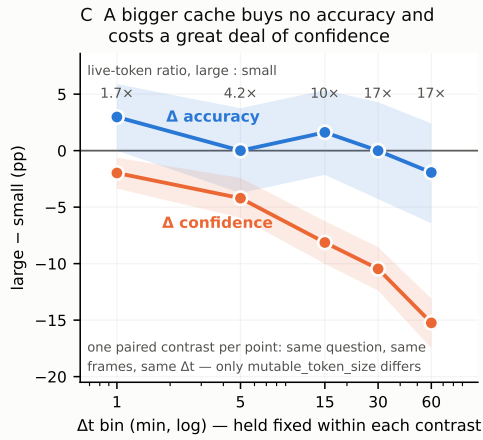
E2 — the positive result is a dose–response

A bigger cache buys no accuracy and costs a great deal of confidence

Δt (min)	Δaccuracy	Δconfidence
1	+3.0 [+0.0, +5.9]	−2.0 [−3.4, −0.6]
5	−0.0 [−3.8, +3.8]	−4.2 [−5.9, −2.4]
15	+1.6 [−2.1, +5.4]	−8.1 [−10.0, −6.2]
30	−0.0 [−4.3, +4.3]	−10.5 [−12.4, −8.5]
60	−1.9 [−6.4, +2.4]	−15.2 [−17.5, −13.1]

A monotone response on one measure with a flat response on the other is much harder to produce by accident than either endpoint alone — and it rules out a threshold artefact.

At $\Delta t = 60$ min a 17× live-token budget (33,684 vs 1,940 live tokens, same frames presented) is worth **−1.9 [−6.4, +2.4]** of accuracy and **−15.2 [−17.5, −13.1]** of confidence.



The pair: the dissociation replicates, and its driver rotates

Differenced contrasts — the same manipulation run in both regimes, paired on the same questions

manipulation, Δconfidence	**E1**	**E2**	difference (**E2** − **E1**)
evidence in context vs none	+3.3	+0.5	−2.8 [−5.1, −0.4]
+45 min elapsed, frames + evidence fixed	−3.7	−0.1	+3.5 [+1.7, +5.4]
3.8× more presented tokens	−0.6	−6.7	−6.1 [−8.1, −4.1]
no-memory vs recall question	+14.3	+12.6	−1.7 [−3.3, −0.1]

Invariant. Accuracy responds to whether the evidence is present (**+9.7 [+4.8, +14.5]** / **+8.1 [+2.9, +13.1]**); confidence does not. Changing the memory architecture did not fix evidence-blindness — it *deepened* it.

Rotated. In **E1** confidence slid with elapsed time and ignored context volume. In **E2** it ignores elapsed time and slides hard with context volume. Both halves of the swap exclude zero.

The third row is the sharpest single number either experiment produced: one manipulation, same frames, same token ratio, and the *accuracy* cost is identical across regimes (**+0.8 [−4.0, +5.4]**) while the *confidence* response differs by **−6.1 [−8.1, −4.1]**. That rules out "**E2** is just a harder task".

What I think this adds up to

One account, no extra assumptions

Confidence is computed from surface properties of the context — how much of it there is, and how long ago it says that was — **not from whether the information needed to answer is present.** When several surface properties are available it latches onto whichever varies; remove one and it moves to another. What it never latches onto is evidence.

It explains, without further assumptions:

- why evidence presence moves accuracy and not confidence in *both* regimes;
- why the driver swaps when the clock is removed;
- why **E1**'s confidence decline is *item-blind* ($+1.0$ $[-2.8, +4.9]$ between items kept and items lost, $p = 0.618$) — a surface property of the whole context cannot distinguish items;
- why the raw Δt sweep looks like forgetting in both regimes and is not.

This is a harder problem than the one we proposed. "Confidence stays flat while accuracy falls" is a failure to update. What we see is confidence moving up to 15 points on variables that predict accuracy *not at all* — a failure of what is being tracked.

The one genuinely new phenomenon in E2

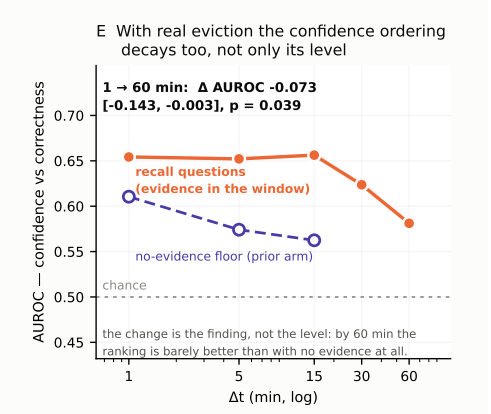
With real eviction, the confidence ranking decays — not only its level

AUROC of confidence against correctness:

	E1	E2
AUROC at $\Delta t = 1$ min	0.643	0.654
AUROC at $\Delta t = 60$ min	0.655	0.581
no-evidence floor	0.568	0.574

Change over the sweep: **E1** $+0.012$ $[-0.046, +0.071]$ (flat) against **E2** -0.073 $[-0.143, -0.003]$, $p = 0.039$.

Why only here: under full-reprocess the evidence is always present at *some* resolution — diluted, never removed. Under a bounded cache it is genuinely gone.



Caveat. One test, interval nearly touching zero. **The change is the finding; the level is not.**

Three readings I want to rule out before we write any of this

1. **"The elapsed-time effect disappeared under streaming."** *No.* EgoStream ingests through the image path, which carries no time markers, so we injected a clock as text — and a separate probe (X1b) established *before* the sweep ran that the model cannot read it once the cache saturates. The axis is **unavailable**, not null. An input the model cannot perceive is not a manipulation that failed.
2. **"Streaming makes the model more overconfident."** *Not supported.* That is a level, and **E2** changes the system prompt and the ingestion path along with the regime. Contrasts are comparable across regimes; levels are not.
3. **"The model tracks the items it actually lost."** *Not yet.* **E2**'s difference between kept and lost items is -3.8 [-8.5, $+1.1$], $p = 0.118$, on 107 lost items — suggestive, underpowered, and the only hint of genuine metamemory anywhere in either experiment.

Calling an **unavailable** axis a **null** is the single most damaging misreading available here — and it is the one a reviewer will make for us if we do not say it first.

Where our own numbers are softer than the headline

From this week's audit of the cache code — found by us, not by a reviewer

issue	what to do about it
The **−15.2 [−17.5, −13.1]** budget headline spans a **change of mechanism**, not just of size — the small-budget arm is the only condition with offload and retrieval in play.	Lead with the offload-free pair: **−6.0 [−7.3, −4.7]** at 3.7× the live tokens, accuracy null throughout. The dose–response *survives* inside one mechanism, which was the most plausible way for the headline to have been an artefact.
"Context length" names **two different variables**: how much the model *holds*, and how much it *threw away*.	Relabel, and use the broader unifier: *confidence falls with the amount of visual input the model has processed, whether or not it retained it.* Both rows fit it; the account gets stronger.
"The evidence is genuinely evicted" is **inferred**, and eviction is content-scored rather than recency-ordered.	Measurable for a few GPU-minutes and one record field: surviving tokens on the evidence frames vs the window mean. It underpins the AUROC result, so it should be measured.

Source: `memo/STREAMING-CACHE.md` §3–§5 — mechanism read from the EgoStream source, numbers recomputed on the committed records.

What this does to the project's original claim

Half 1 — "a widening gap as evidence ages"

Rejected. Twice, in two regimes, including the one where evidence is truly evicted. Retire it.

Half 2 — "a Δt/evidence-conditioned recalibration recovers most of the reliability at near-zero cost"

Needs revision on both variables.

The corrected version

- Δt is the *wrong conditioner*. It worked in **E1** by accident; in **E2** the model cannot even read it. The variable that actually drives confidence is `live_tokens` — observable at inference time, and free.
- "Evidence-conditioned" is *not implementable*: it presumes knowing whether the evidence is in context, which is exactly what the model cannot report.
- **The headroom differs by regime.** Recalibrating a level only works if the ranking survives: it does in **E1** (AUROC flat), and it does not in **E2** (-0.073 $[-0.143, -0.003]$).

The last point is the most consequential thing the pair says, and it is only visible because both regimes were run: **the cheap fix looks cheaper the less realistic the memory system is.**

Part 2

Is the confidence probe the right construct?

Should confidence be a softmax over the four option letters,
or over the whole vocabulary? Answered on the committed records, no GPU.

The question, and the answer

The alternative was already on disk — both backends log the full-vocabulary logprobs at the letter ids

With M = the next-token mass on {A,B,C,D}, the vocabulary-normalised confidence is exactly $c_{\text{vocab}} = c_{\text{restricted}} \times M$, recoverable for every one of the 5,540 + 16,565 committed records at **zero GPU cost**. No re-run was needed and none was done.

Verdict: keep `max softmax({A,B,C,D})` as the primary metric.

- It is the *coherent denominator* — correctness is argmax over those four letters by construction, so ECE against MCQ correctness is only coherent for the restricted version.
- It is *field-standard*, not our idiosyncrasy: EgoSAT Eq. 6 is our definition verbatim, tracing to Tsvilodub et al. 2024.
- The alternative changes nothing we report — except one number, where it changes it for the wrong reason.

Source: `memo/CONFIDENCE-PROBE.md`

The design choice that actually deserves scrutiny is not the denominator.

It is **option-letter bias**, and a second quantity the restricted softmax throws away turns out to carry a real signal. Both are on the next slides — and the second is the cheapest positive result available to us.

Everything here was recomputed with the *unchanged* contrast machinery that produced the two authoritative artifacts, so it is directly comparable to the headline numbers.

Where the choice makes no difference — which is nearly everywhere

The model puts essentially all its mass on the letters.

	E1	E2
mean mass on {A,B,C,D}	0.99988	0.99530
argmax disagreements: **0 of 23,687** records		

Prediction — and therefore *accuracy* — is identical under the two definitions in every record we have.

Every contrast in both experiments is unchanged to 0.01 pp, including the two headlines: **E2**'s context-length effect and its null on elapsed time at fixed frames.

And the ranking does not care how you collapse the four-way distribution to a scalar.

AUROC against correctness, memory rows:

discriminator	E1	E2
max-p, restricted (as shipped)	0.6276	0.6199
max-p, vocabulary-normalised	0.6275	0.6203
negative entropy	0.6228	0.6215
top1 − top2 margin	0.6254	0.6171

The stronger robustness statement: **however the distribution is collapsed to a scalar, the answer is the same to 2–3 decimals.** No conclusion rests on choosing max over entropy or margin.

The one place it does matter — and it would silently rewrite a verdict

evidence in context vs none, Δconfidence	E1	E2
restricted (as shipped)	+3.32 [+1.24, +5.40]	+0.55 [−1.52, +2.65]
vocabulary-normalised	+3.32 [+1.24, +5.40]	+2.61 [+0.33, +4.94]

Under vocabulary normalisation **E2**'s confidence *stops* being evidence-blind. Three things we have written down would change:

- the streaming module's verdict selector flips from *"the dissociation replicates"* to *"confidence tracks eviction"* — it reads that interval's sign directly;
- the cross-regime claim that confidence's evidence response is *weaker* under streaming (−2.8 [−5.1, −0.4]) loses significance;
- our position in the EgoSAT comparison moves from the extreme band to the middle of it.

So: do not switch the denominator quietly — it is not a cosmetic normalisation. The fix is to make the sensitivity visible in the artifact: assert the denominator in the module, or emit both values alongside the verdict.

What the flip is actually made of — and it is a result, not a nuisance

Implicit abstention: with a bounded cache and no evidence, the model sometimes stops wanting to emit a letter at all

The effect is not a smooth rescaling — flooring the letter mass at 0.90 removes it ($+1.18$ [-0.92, $+3.28$], p $= 0.259$). It rides on a small, heavy-tailed minority of rows:

- 129 rows (1.38%), over 79 of 373 questions carry more than 10 % off-letter mass;
- 8.76% of *no-evidence* rows, against 0.42% of evidence rows;
- none at all at $\Delta t = 30$ or 60 min;
- in **E1** the maximum over every record is 0.0034 — **the phenomenon does not exist there.** It is specific to the bounded cache.

Paired within-question, on *off-letter* mass (positive = more hedging):

contrast	**E2** Δoff-letter
evidence removed	-3.01 [-4.40, -1.82]
raw Δt sweep	$+0.60$ [$+0.21$, $+1.11$]

This converts one of E2's nulls into a measured effect — a metacognitive signal living exactly where max-p is silent, at no GPU cost.

Report it as **abstention**, not confidence: the two channels move in *opposite* directions across Δt, and folding them lets a 1.4 % tail rewrite the previous slide's verdict.

The construct threat that is real — option-letter bias

	gold letters				what the model predicts			
E1	A .222	B .256	C .261	D .261	A .124	B .280	**C .435**	D .161
E2	A .222	B .256	C .261	D .261	A .134	**B .342**	C .330	D .194

In **E1** that is 21.4% on gold-A vs 58.6% on gold-C. With *no evidence in the window at all*, the model still predicts the same skewed letters at mean confidence 0.659 (**E1**) and 0.803 (**E2**).

This is good news twice over.

- It is a *mechanism* for evidence-blind confidence, which we currently report only phenomenologically. A content-independent option prior does not move when evidence is removed.
- **The paired design largely survives it.** The letter prior is a property of the question, held fixed across arms, so it differences out of every within-question contrast to first order.

It will be asked, and EgoSAT already runs the check we do not (their option-order re-shuffle).

Cheap strengthener: a **cyclic option-permutation arm** at one Δt bin on the evidence and no-evidence arms, averaging per-option probability across permutations before the softmax. 4× cost on a subset.

It is also a further reason the level claims are weak and the contrasts are not — which is already our stated reading.

Part 3

EgoSAT: collision and cross-validation

Prior art, not a scoop. It costs us one novelty claim and independently replicates our central finding.

What EgoSAT is, and what it costs us

Lei et al., arXiv 2606.24422, 23 Jun 2026

A streaming egocentric benchmark: 1,997 Ego4D sessions, 165 h, ∼4,800 MCQ, six tasks spanning retrospective / present / prospective under strict online prefix truncation, plus an answerability model for future events and a training-free ROI-based KV selection method.

Four places we actually collide:

- **The confidence probe.** Their Eq. 6 is our definition verbatim. Hard collision — cite and move on.
- **Confidence vs temporal distance.** Same estimand, non-overlapping scale (8 / 16 / 24 s against our 1–60 min).
- **Answerability-conditioned confidence.** Structurally our evidence contrast — and it replicates.
- **Online vs offline.** Same question as our regime pair, but cross-*model*, not within-backbone.

Must fix in our framing

- Drop any *"first to evaluate confidence in egocentric streaming"* claim — their appendix explicitly claims that column.
- Attribute the option-letter probe to Tsvilodub et al. 2024 / EgoSAT Eq. 6 rather than presenting it as our design decision.
- Retire "confidently wrong" as phrasing — it is their abstract's term.
- Add EgoSAT and the six works it surfaces to our citation index.

Position instead on **evidence age** and the **elapsed-time-vs-context-volume decomposition**, which nobody in this literature runs.

The win: our central finding replicates outside our lab

Their answerability split, our evidence split — different dataset, different backbones, same shape

model / experiment	Δaccuracy	Δconfidence	under-reaction
Qwen2.5-VL-72B	+9.45 / −4.57		sign-flipped
Qwen2.5-VL-7B	+10.05 / +1.75		5.7×
TimeChat-Online-7B *(streaming)*	+4.51 / +0.80		5.6×
ROI-TimeChat-Online *(streaming)*	+5.42 / +0.74		7.3×
ours — E1 (full-reprocess)	**+9.7 [+4.8, +14.5]**	**+3.3 [+1.2, +5.4]**	2.9×
ours — E2 (streaming)	**+8.1 [+2.9, +13.1]**	**+0.5 [−1.5, +2.6]**	CI through zero

Three things line up: the **accuracy** effect band (theirs +4.5 to +11.6 pp, ours +9.7 [+4.8, +14.5] and +8.1 [+2.9, +13.1]); the **confidence under-reaction** (theirs 3–7×, or an outright sign flip); and **streaming is the worse case in both** — their two native streaming models separate correct from wrong answers by only +2.78 and −0.35 confidence points, which is our AUROC decay in a coarser form.

Our headline is no longer a one-dataset, one-model result. It should be cited as convergent evidence in exactly those terms.

Caveat to state rather than elide: their split is over intrinsic *future predictability*, ours over presence of *past evidence* — the same claim only if "answerability" is accepted as the shared latent.

Where they look like they contradict us — and why they do not

1. Confidence does not fall with distance in their table (six of eight ≥ 0 slopes).

*Against **E2**: agreement.* Our streaming result is flat (-0.1 [-1.7, $+1.4$]); their two streaming models are flat-to-positive ($+0.99$ and $+4.61$). Both say: under a real bounded cache, confidence does not fall with retrospective distance.

*Against **E1**: apparent disagreement*, for three reasons that are all estimand differences — 45 min vs 16 s; their slope is fitted only on items where *all three probes are correct*; and their point estimates carry no intervals.

2. Their accuracy declines with distance (-2.6 to -7.4 pp over $8 \rightarrow 24$ s) **where our token-matched arms say elapsed time costs no accuracy.**

Their decline over *16 seconds* is the size of our entire raw $1 \rightarrow 60$ min sweep. Their lag axis is not controlled for context volume or distractor difficulty — and their own *streaming* models, whose caches saturate so context stops growing, show the smallest or non-monotone declines. That is what our account predicts.

The defensible framing, in one sentence: their uncontrolled lag axis reproduces our uncontrolled raw sweep — and we have the control that says what it is made of (**E1**: the accuracy decline is **-2.4 pp of context** against **$+0.3$ pp of elapsed time**; **E2**: the confidence decline is **-6.71 pp of context** of an observed -6.82, residual $+0.02$).

What keeps its value

Ranked by how much I would defend each in review

1. **The token-matched decomposition.** Nobody in this literature separates elapsed time from context volume, and the previous slide shows the omission is load-bearing for a published conclusion.
2. **Within-backbone streaming vs full-reprocess.** Their online-vs-offline comparison differs in backbone, training data and tuning at once, and cannot support the causal claim it makes; ours is a plug-in memory manager over frozen weights.
3. **Time scale.** 1–60 min of gap-aware *stream* time on a 44 h life-log against their 8–24 s in 5-minute clips — 40–450× the horizon, and the regime a life-logging assistant actually operates in.
4. **Statistics.** Paired within-question contrasts, cluster bootstrap, decomposition-residual assertions, AUROC and ECE. Their tables are point estimates with no intervals and no pairing.
5. **A token-matched no-evidence floor.** Their blind baseline removes vision entirely, changing the token budget and the modality at once; ours removes only the evidence.
6. **Confidence on the memory-budget axis.** They vary the memory horizon and report *accuracy only* (3.6 pp and 2.4 pp, both non-monotone) — our **−15.2 [−17.5, −13.1]** confidence response is exactly the cell they leave empty.

Their SFT result — that fine-tuning improves calibration mainly by suppressing overconfident errors — is supporting evidence for our recalibration angle, not competition for it.

Part 4

What I propose next

Four of the next five things cost under a GPU-hour.

Ranked by what they change per GPU-hour

Four of these six cost under a GPU-hour, and two cost none

experiment	cost	what it settles
Strip the timestamps from E1 and re-score the two token-matched cells	≈1 GPU-h	Whether **E1**'s elapsed-time effect is a response to printed time tokens *at all*. If it is, the story stops depending on the memory architecture — our most confounded claim.
Recalibrate on `live_tokens`, per regime	free	The corrected Half-2 claim. The AUROC asymmetry predicts it succeeds in **E1** and partially fails in **E2** — sharp either way.
Off-letter mass as a reported quantity, plus a module regenerating this week's tables	free, CPU	Turns the abstention signal into a result, and puts the denominator sensitivity in an artifact rather than a memo.
Re-probe the injected clock in the native format	minutes	Whether "cannot read the clock" is a property of the cache or of the format we injected.
Cyclic option-permutation arm at one Δt bin	4×, one bin	The option-letter bias check EgoSAT runs and we do not.
Explicit abstention (a fifth option)	∼2.4 GPU-h	Given the words to say "not enough information", does the model use them when the evidence is gone?

Decisions I would like from you

1. **Retire the original hypothesis?** It has now failed in both regimes, the second of which was the setting it most deserved. My recommendation is to retire it and lead with the dissociation and its rotation.
2. **Which follow-up first?** My order is: timestamps-off on **E1**, then recalibration on `live_tokens`. The first is the cheapest thing that could change the story; the second is the corrected version of what we promised.
3. **How hard to position against EgoSAT?** They hold the "confidence under streaming" flag; we hold the decomposition, the within-backbone comparison, the time scale and the statistics. I would cite them as convergent evidence in the introduction rather than fight for the framing.
4. **Is the abstention channel worth a section?** It is free, it is a measured effect where we currently report a null, and it is the only genuinely metacognitive signal in either experiment — but it is a 1.4 % tail and it needs the top-k dump to say what those tokens are.

Part 5

Backup

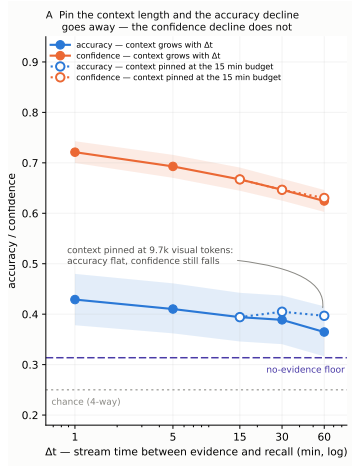
The raw material behind each claim.

Backup — E1, the raw Δt sweep

Confounded by construction: the window grows with Δt. This is the material the decomposition consumes, not a result.

Δt (min)	acc %	conf %	gap	visual tokens
1	42.9	72.1	+29.2	1,279
5	41.0	69.3	+28.3	3,679
15	39.4	66.7	+27.3	9,679
30	38.9	64.7	+25.8	18,679
60	36.5	62.4	+26.0	36,679

The usable dynamic range is narrow. Bin-1 accuracy 42.9 % sits 11.5 pp above the no-evidence floor of 31.4 %, and chance is 25 %. **E2**'s band is the same width (11.8 pp). Any statement about an accuracy *level* is made inside that band — which is why every headline here is a contrast.



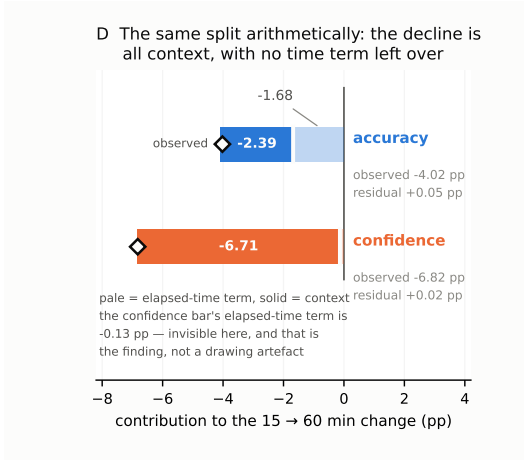
Backup — the decomposition, in both regimes

The one arithmetic that makes the raw sweep interpretable

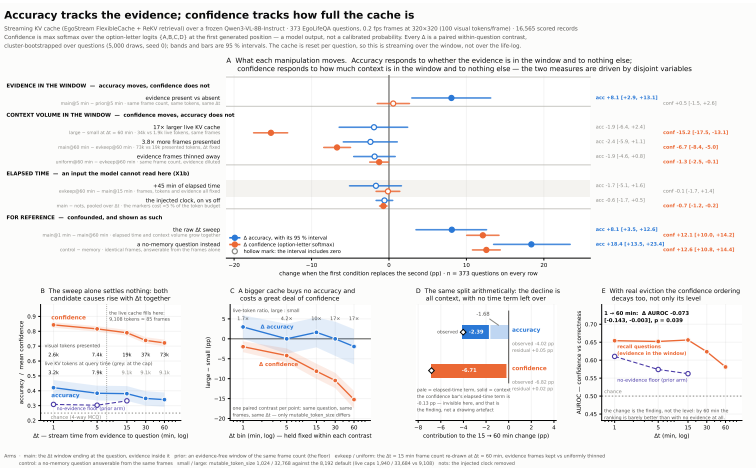
E1: contribution to Δt 15 → 60 min	acc	conf
elapsed-time term *(token-matched)*	+0.3 pp	−3.7 pp
context-length term *(same Δt)*	−2.4 pp	−0.3 pp

Accuracy's decline is entirely the context term; confidence's is almost entirely the elapsed-time term. The raw sweep is a composite of two effects pointing in different directions for the two measures — which is why reading it as "forgetting" is wrong.

In **E2** (right) the same code gives **−6.71 pp** of context against **−0.13 pp** of elapsed time, observed −6.82 pp, residual +0.02 pp. The elapsed-time term is not merely small — it is absent.



Backup — E2, the whole headline figure



Backup — E2, the streaming contrasts in full

manipulation	Δaccuracy	Δconfidence
evidence in context vs none	+8.1 [+2.9, +13.1]	+0.5 [−1.5, +2.6]
+45 min elapsed, frames and evidence fixed	−1.7 [−5.1, +1.6]	−0.1 [−1.7, +1.4]
3.8× more presented tokens	−2.4 [−5.9, +1.1]	−6.7 [−8.4, −5.0]
3.7× live-token budget (offload-free pair)	+1.8 [−0.4, +4.0]	−5.4 [−6.5, −4.4]
evidence frames thinned away	−1.9 [−4.6, +0.8]	−1.3 [−2.5, −0.1]
injected clock on vs off	−0.6 [−1.7, +0.5]	−0.7 [−1.2, −0.2]
raw Δt sweep *(confounded)*	+8.1 [+3.5, +12.6]	+12.1 [+10.0, +14.2]
no-memory vs recall question *(between-items)*	+18.4 [+13.5, +23.4]	+12.6 [+10.8, +14.4]

All 373 questions on every row, paired within-question, cluster-bootstrapped over questions (5,000 draws, seed 0).

Source: `artifacts/x9_streaming.md`, generated from `x9_regime_comparison.json`.

Backup — what the cache actually holds

Why E2 is a different experiment and not just a slower one

The cache is split into a pinned priming region, an evictable mutable region (the experiment's budget knob), and a protected hat holding the last two frames. On overflow the pruning pipeline drops the lowest-value and most redundant tokens.

Two consequences that are not visible from the config:

- **Eviction is content-scored and global, not recency-ordered.** "Older evidence is pushed out first" is not this policy — so a flat accuracy-vs-Δt response is what this cache *predicts*.
- **Eviction is permanent in the primary condition.** Audited over every streaming record: 0 of 12,890 default and large rows offloaded anything, against 3,675 of 3,675 small rows.

The sweep crosses saturation at ≈85 frames (≈7 min). By $\Delta t = 60$ min only ≈11% of the presented tokens survive, and the live token count is pinned at 9,108 in every bin from 15 min on.

And **E1**'s clock is *text* the model tokenizes (<150.0 seconds>), not a positional encoding — confirmed by decoding the prompt through **E1**'s own input builder. That is why stripping it is a real experiment and a cheap one.

Source: `memo/STREAMING-CACHE.md` §2, §6–§7 — mechanism read from source, numbers recomputed from the committed records.

Backup — how this deck is built

Same convention as the experiments: modules separated along verifiable-artifact boundaries

module	artifact	what checks it
B1 `b1_numbers.py`	`generated/numbers.tex`	Re-derives all **128** macros from the metrics JSON; memo-sourced values carry a literal that must still appear in the memo.
B2 `b2_figures.py`	`figures/*.pdf`	Panels are crops, never redraws; manifest pins the source figures by sha256 and the crop boxes.
B3 `preamble.tex`	the theme	—
B4 `slides/*.tex`	one file per agenda item	B5 compiles each file on its own, so a section can be edited without the rest.
B5 `b5_verify.py`	the built PDF	Nine checks; fails on a stale generator, a changed source figure, a missing figure, an undefined macro, a LaTeX error, or a visibly overfull slide.

No number on any slide was typed by hand. Each one is a macro whose value and provenance are recorded in `generated/numbers.json`; if an experiment is re-run, `make` rebuilds the deck and `make verify` fails until it is rebuilt.

Precedence, as in the repo: `m8_teaser.md` and `x9_streaming.md` are authoritative for numbers; `INTERPRETATION.md` governs how to read them; this deck is interpretation on top of both.